EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

An Earth observation that survives an agent handoff.

Agent A cites a satellite observation. What can agent B check without trusting A or A's model?
emem makes satellite observations, raw or derived, **addressable by place, band and time**. Each is a record named by the **BLAKE3 hash of its bytes** and signed in a batch. The exact evidence A cites survives a change of model and a **handoff**; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name or signature.
Trace token · record bytes · cell · pixel · receipt · log · key

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode in AI's reasoning with emem.

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From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor	Location · product · time	Canonical record + CID	Pass the reference	Retrieve the same record	Hash · bind · sign · source	Reason with cited evidence
Keep the source scene.	Lookup identity	Batch attestation	MCP · REST · A2A	Recover its provenance.	**Policy: refuse if one fails.**	Carry the reference onward.

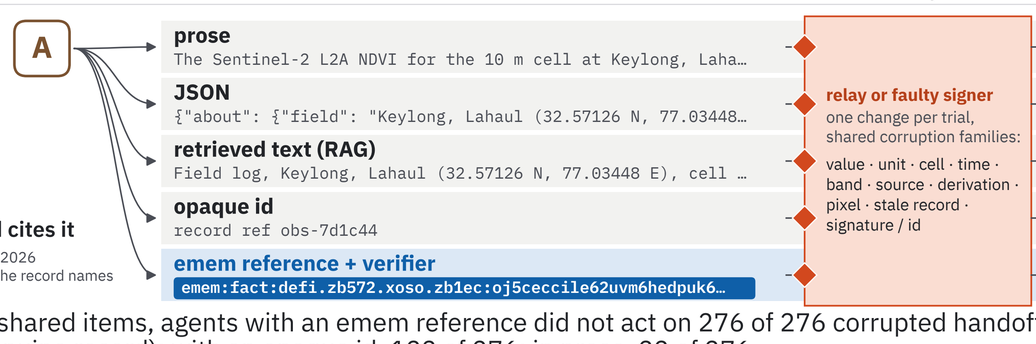
1 Does agent B act on a corrupted handoff?

The same corruptions, five handoff conditions.

Agent A cites a satellite reading. A relay or a faulty signer changes the evidence. Agent B acts on it, declines it or uses the genuine record.



Agent A reads NDVI 0.4709 and cites it
Sentinel-2A L2A, 25 Sep 2026
outlined: the 10 m pixel the record names
RGB from recorded grids



Agent A cites	handoff	relay	Agent B	B did not act on corrupted evidence	declined	used the genuine record
				agents, pooled Claude		
prose	The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha…	relay or faulty signer	reads the text	22 / 276	22	0
JSON	{"about": {"field": "Keylong, Lahaul (32.57126 N, 77.03448…	one change per trial, shared corruption families: value · unit · cell · time · band · source · derivation · pixel · stale record · signature / id	reads the fields	112 / 276	106	6
retrieved text (RAG)	Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell …		retrieves a passage	56 / 276	56	0
opaque id	record ref obs-7d1c44		fetches the sender's record	134 / 288	88	46
emem reference + verifier	emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk6…		resolve · re-hash · place / time / band · signature · log · recompute · source re-read	300 / 300	264	36

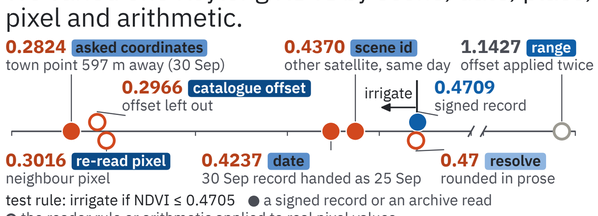
Across the 23 shared items, agents with an emem reference did not act on 276 of 276 corrupted handoffs (240 declined, 36 used the genuine record); with an opaque id, 122 of 276; in prose, 22 of 276.
By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked, so no source re-read (SAT-042 harness, not flown).

Figure: Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 pooled, full item set per condition. 2,794 scored trials (1 Oct 2026, two constructed tasks, Qwen included) span 25 corruption items, 2 controls and 1 exploratory item. The emem lane adds a verifier and an instruction, so the figure tests the workflow. **Matched baseline** (4 Oct): emem's checks minus hash, signature and log, run on the JSON, accepted 8 of 23 corruptions (3 with four more checks; emem's verifier 0); in an exploratory run, one trial each, instructed Haiku agents given those checks acted on the same 8. Qwen2.5-7B acted on 10 of 25 corruptions behind a fail-closed resolver (2 of 25 when told to check): gate the action outside the model.

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



test rule: irrigate if NDVI ≤ 0.4705 ● a signed record or an archive read ○ the reader rule or arithmetic applied to real pixel values

Six cross the constructed irrigation threshold; one lies outside the valid NDVI range. Each calls for a different check.

3 Why can a value change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \varepsilon$$

Δz: change in readout. The terms name possible causes (environment, sensor, geolocation, encoder, residual); we have not measured their shares.

7 What does a checked reference establish?

The record, not physical truth

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token · **record** · **source pixel**

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG[row, col] = DN(b) & Ed25519(σ) & leaf in STH & $STH_{geo.qa}$ prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links (17 checks) in 17.9 s, all verified**. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks… · still trusted: BLAKE3 and the verifier's code
4-5 cell binding hashed cell = token cell = the lat/lng the derivation used · the cell rule
6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code
8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · this STAC item carries no checksum
10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right
12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the signed tree head (STH) geo.qa co-signed · geo.qa is also run by Vortx AI
15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

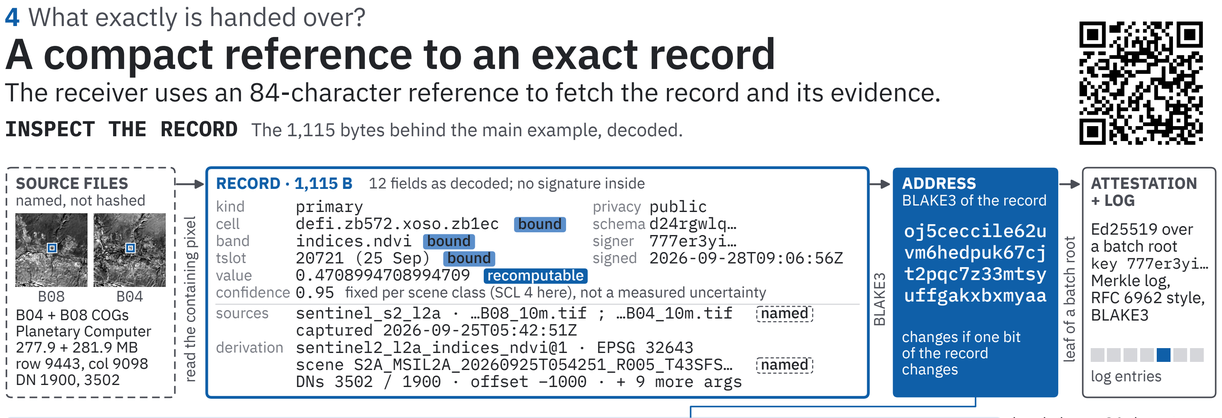
Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 5).

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



`emem:fact:defi.zb572.xoso.zb1ec:oj5ceccile62uvm6hedpuk67cjt2pqc7z33mtsyuffgakxbxmyaa` handed over: 84 characters, 46 GPT-4 tokens

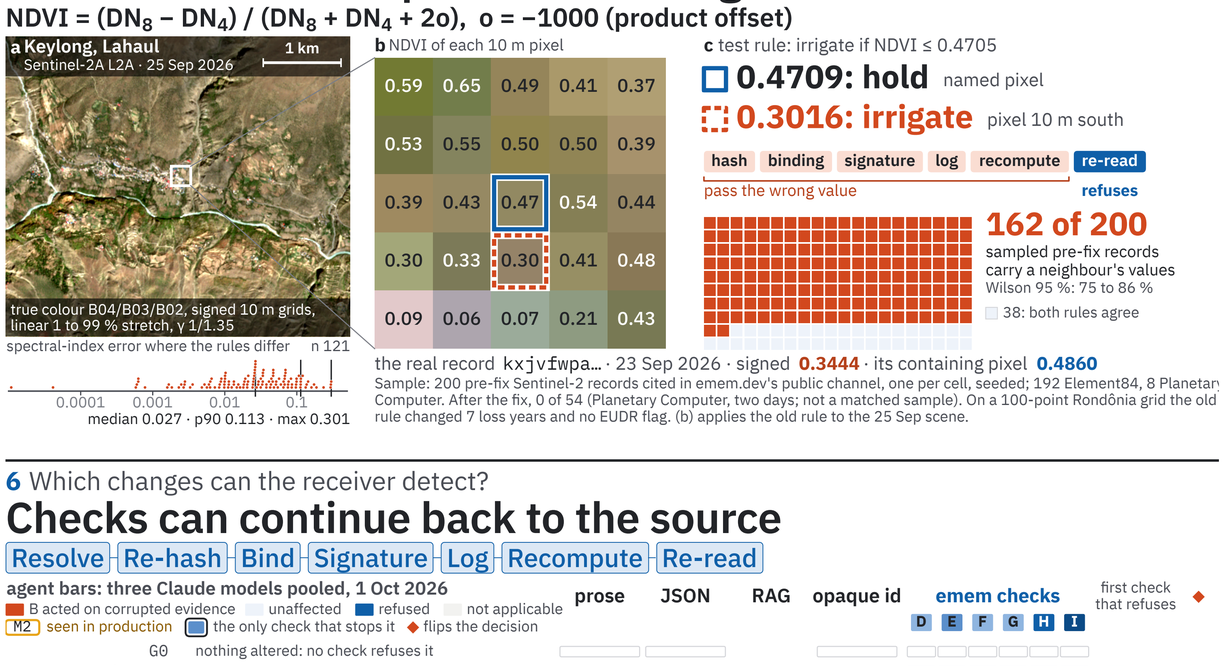
BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE RECORD, AS CODED · fact.rs, cbor.rs, geo.rs

```
fact   f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation{fn_key, args}, privacy, schema_cid, signer, signed_at)
name   b = CBOR(f), declaration order, NaN as f9 7e00, −0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem:fact:<cell>:<fact_cid>
cell   lat_q = round((lat+90)/180·(2²¹−1)), lng_q = round((lng+180)/360·(2²²−1)); pitch 8.583e-5° = 9.52 × 8.06 m here
```

5 A valid record can preserve a wrong source read

NDVI = (DN₈ − DN₄) / (DN₈ + DN₄ + 2o), o = −1000 (product offset)



Sample: 200 pre-fix Sentinel-2 records cited in emem.dev's public channel, one per cell, seeded; 192 Element84, 8 Planetary Computer. After the fix, 0 of 54 (Planetary Computer, two days; not a matched sample). On a 100-point Rondônia grid the old rule changed 7 loss years and no EUDR flag. (b) applies the old rule to the 25 Sep scene.

6 Which changes can the receiver detect?

Checks can continue back to the source

Resolve · Re-hash · Bind · Signature · Log · Recompute · Re-read

agent bars: three Claude models pooled, 1 Oct 2026
B acted on corrupted evidence · unaffected · refused · not applicable
M2 seen in production · the only check that stops it · ◆ flips the decision

			prose	JSON	RAG	opaque id	emem checks	first check that refuses
	G0	nothing altered: no check refuses it						
value	M1	stated value changed in its last bit	0/12	0/12	0/12	3/12	C	◆
	M2	stated value rounded to "0.47"	6/12	6/12	1/12	12/12	C	◆
	M8	value 0.45, re-encoded and re-hashed	0/12	6/12	0/12	6/12	F	◆
cell	M4	record cited for another cell	5/12	12/12	6/12	6/12	E	
	M9	cell changed inside the record, re-hashed	5/12	1/12	6/12	6/12	F	
time	M10	tslot changed to look current, re-hashed	0/12	6/12	6/12	6/12	F	
band	M6	a record for another band handed over	0/12	12/12	6/12	10/12	E	
source	M11	source scene id changed, re-hashed	0/12	0/12	0/12	0/12	F	
derivation	M12	offset changed, value recomputed, re-hashed	0/12	1/12	0/12	4/12	F	◆
	M14	value (0.46) disagrees with the signed DNs	0/12	6/12	0/12	6/12	H	◆
stale / current	M5	an older record handed over as current	0/12	11/12	6/12	11/12	E	
	M16	a second signed version shown only to B	0/12	6/12	6/12	6/12	G	
signature / id	M3	last bit changed inside the served bytes	0/12	0/12	0/12	0/12	D	
	M7	token miscopied by one character	n/a	n/a	n/a	12/12	D	C
	M13	value 0.45 under the forger's own key	0/12	6/12	0/12	6/12	F	◆
source pixel	M15	DNs read from the pixel 10 m south	0/12	0/12	0/12	0/12	I	◆
entity	M17	same record; A meant another entity (out of scope)					none	

B did not act on corrupted evidence	prose	JSON	RAG	opaque id	emem checks
deterministic receiver, same corruptions	22 / 276	112 / 276	56 / 276	134 / 288	300 / 300
	0/15	0/15	3/16		16/16

C resolve · D hash · E binding · F signature · G log · H recompute · I re-read · D to I are cumulative: each adds one check

Bars: B did not act on it (declined, or used the genuine record), n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: deterministic ceiling. Totals pool 23 to 25 items, 1 Oct 20 Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Agents with an emem reference decided correctly on 71 of 72 genuine controls and declined one.

JSON and an opaque id caught some changes from their fields. On the live pre-fix record, agents acted 4 of 4 times uninstructed, 2 of 4 with resolve-and-bind instructions and 0 of 4 with a source re-read.

8 Can a later agent recover the earlier state?

Revisit the earlier evidence

C_τ = TxAsOf(O*, a, b, τ)
recall = $\max_{(t, cid)}$ {O in C_τ : t ≤ t*}

Latest-as-of mode: take the versions signed by τ (signed_at, the signer's clock), then the latest valid time ≤ t*. If no candidate exists, recall returns an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

same place + band

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90	Copernicus DEM30
earlier CID	later CID

as of 15 Jun: 918.0 m

The next agent can recover the earlier citation.
The later 915.07 m reflects a provider change.

9 What does the memory contain?

Observations + temporal edges

M = (O*, E*) O = (a, b, t, v, u, p, s)

a: location cell · b: variable · t: valid time · v: value
u: uncertainty, if recorded · p: provenance, recipe, signed_at
s: attestation associated with the observation

E = (subject, predicate, object, t_{from}, t_{to})

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t); a cid names a record, not an object
cid(record) = BLAKE3(CBOR(record))

Use it in an EO workflow

Compare dates · recall, diff
Assemble evidence · merge, trace
Retain conflicts · competing, evolve
Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting estimates and pass their references on. A request reuses stored records and fetches and signs missing ones from the archive.

10 What else can an agent carry?

Beyond a single observation

In emem, the token family is how agents read Earth observation: one grammar for a place, a value, a raster, a time series or a device run.

data need	grammar, after emem:	what its id hashes
a place	`cell:<cell64>`	the quantised lat/lng; no hash
one value	`fact:<cell>:<cid>`	BLAKE3(CBOR(fact)), 52 chars
nothing there	`fact: · kind absence`	same; + BLAKE3(reason)[:16]
an embedding	`fact: · 128 / 1024 floats`	same; not re-runnable
a derived value	`fact: · op, parents`	same; pure ops re-runnable
a raster	`raster:<a>:<b>:<t>:<d>`	fact d names BLAKE3(grid)
a raster in time	`cube:<a>:<b>:<t0..t1>:<d>`	BLAKE3(CBOR([member d…]))
a raster set	`rasterset:<set>:<d>`	BLAKE3(CBOR([d…, purpose]))
many facts, one id	`bundle:<cid>`	BLAKE3(p, cell\|band\|t\|cid)[:16]
a file, one chunk	`tree:<file>#row=i`	BLAKE3(note)[:16]; root inside
a reasoning stage	`state:<cid>`	BLAKE3(CBOR(state record))
a named object	`entity:<cid>`	BLAKE3(anchor)[:16]: a label
a device run	`trace:<cid>`	BLAKE3(CBOR(signed trace))
an evidence chain	`track.v1 note`	link = BLAKE3(prev, ref)[:16]

14 data needs; embedding and absence share the fact grammar.
Names, sets and records hash differently.
Grey: encoders retired; device runs in a reference harness.

One bundle resolves to eight facts; four reasoning-state addresses re-hash. Neither check validates an agent's reasoning.

11 Satellite execution evidence

A signed trace binds the reported run

SAT-042 scripted pass · reference harness · 30 Sep 2026.

Enrol	Write, no trace	Capture the pass	Unbound fact	Bound batch	Score vs anchor	Rewrite one log
ENROLLED	REFUSED	SIGNED	REFUSED	ADMITTED	SCORED	REFUSED

8 trace layers, each log digest chained; seq 2 rewritten after signing: broken at seq 3

syscall seq 0 · scheduler seq 1 · memory seq 2 ⊗ · sensor bus seq 3 · signal seq 4 · energy seq 5 · thermal seq 6 · storage seq 7

Reference harness, no spacecraft enrolled. The trace shows which key signed the run and that no logged segment changed; it does not show which code ran on which inputs. The gate binds output digests, not band, cell or time.

12 Use emem. Carry the evidence forward.

ChatGPT · @emem plugin
"Signed facts about real places"
Enable emem; paste a token.

Claude · Directory plugin / MCP
"hand another agent a token it can check offline"
Resolve from chat or terminal.

Dify · Marketplace plugin
"Shared memory for AI agents. One address per fact, one signature you check."
Workflow templates, including EUDR

GitHub MCP Registry · Server listing
"Satellites for AI."
Install MCP server

ClawHub · OpenClaw plugin listing
"Shared, signed and content-addressed memory of the physical world…"
Install the OpenClaw plugin.

MORE CLIENTS
Cursor · VS Code · Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript · Docker
LangGraph memory · GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex · AutoGen
CrewAI · Mastra · Agno (see setup notes)

DISCOVERY / MIRRORS
MCP Registry · MuleSoft Exchange · Glama · Smithery
Context7 · Hugging Face (older mirror)

CONNECT
Setup · EO tasks
Reproduce

Measurements: 29 Sep to 1 Oct 2026. · Methods link each figure to its records, data and checks (Connect). · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)